External force · Frayer & doodle notes

Name _____ Partner _____

SECTION A BUILD THE WORD

Focus word External force

From Section A: Object, External force, Force-arrow scale, Net force. For a different word, use the second box to explain how you recognize it.

Meaning in my words Explain it without using the term.	**How can you tell?** Draw a boundary around the lamp alone. What makes the rope's pull external?
Example or sketch Draw and label a case that fits.	**Non-example and why** Show a near miss. What rules it out?

SECTION B DOODLE THE IDEA

Sketch the lamp, its boundary, and its two force arrows. Make their lengths predict upward acceleration.

Chosen object: _____

My marks mean _____

On assignment: External force

SECTION E / BACK / FINAL MEETING ONLY Name _____

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Systems engineer**

Choose the boundary of the simulated object. Draw only the force layers you would send to the acceleration calculation.

☐ **Animation director**

Make a two-layer storyboard: physical forces on XJ, and motion. Tell the animator which Earth arrow belongs in another shot.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor’s check: Distinguish the net-force direction from a possible velocity direction.